The Koide Identity in a Z_3-Symmetric Mass Parametrisation

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Abstract

This paper gives a proof that the selected Z-three-symmetric Brannen mass parametrisation satisfies the Koide identity for every permitted phase and mass scale. That conditional trigonometric identity is the paper's mathematical result; it was previously known in the parametrisation literature and does not establish a physical lepton mechanism.

The trefoil/vortex interpretation and mode-stability discussion are hypotheses. The reported phase and mass scale are obtained by a simultaneous least-squares fit to all three charged-lepton masses, not a prediction of an independent third mass. No physical derivation of the phase, particle mapping, or vacuum model is provided here.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, Koide formula, lepton masses, Z3 symmetry

Project Context

This manuscript constitutes Part I (Paper 02) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 01 of this series [6].

Symbol Conventions

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C)\cdot S\cdot M_s$ or $(\Phi/C)\cdot S\cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

In 1981, Yoshio Koide noticed that the masses of the charged leptons satisfy a remarkably tight relation [4]:

$$Q \;=\; \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \;=\; \frac{2}{3}\,. \tag{1}$$

Using current PDG values [8], $Q = 0.666661$, which agrees with $2/3$ to one part in 10^4. More strikingly, when Koide first wrote the formula, only m_e and m_μ were well-measured. He *predicted* $m_\tau = 1776.97$ MeV. The measured value is 1776.86 MeV [8] — a prediction accurate to 0.006%.

Despite its precision, Eq. (1) has no derivation within the Standard Model. Quartic fermion operators that might enforce it violate renormalisability. No symmetry of the Standard Model selects $Q = 2/3$ over any other value. The formula is accepted as a numerical coincidence.

CDFT offers a different starting point. In Paper I of this series [6], we showed that the electron is a stable vortex ring in a transport medium vacuum, with aspect ratio $\chi = R/a = 1/\alpha = 137.036$, where α is the fine-structure constant. That vortex has a U(1) rotational symmetry around its axis. The present paper asks: what happens when that symmetry is spontaneously broken by an azimuthal perturbation?

The answer, as we show in Section 3, is that the vortex locks into a $\mathbb{Z}_3$-symmetric trefoil. And as we prove in Section 4, any $\mathbb{Z}_3$-symmetric mass triplet satisfies the Koide formula exactly. Within this construction, the Koide formula is a theorem about rotational symmetry rather than a fitted numerical coincidence.

1.1 Mujjabi triadic stability principle

Paper I states the capacity law and the stability attractor for the ground-state vortex. This paper states the first symmetry consequence: when the regulator admits stable azimuthal excitation, the first nontrivial locked mode is triadic. In CDFT terms, the electron, muon, and tau are treated as three phase-regulated expressions of one capacity-limited vortex family, not as three unrelated mass labels. We call this the Mujjabi triadic stability principle:

> The first stable generation structure of the CDFT regulator is the $\mathbb{Z}_3$ trefoil, and its mass bookkeeping is forced into the Koide invariant.

The principle is mathematical inside the stated vortex-mode model. Its physical status still depends on the later topology, density, and experimental tests.

2 Recap: the CDFT Vortex Ring (Paper I)

We summarise the results of Paper I that are needed here. Throughout this recap, α denotes the electromagnetic fine-structure constant, $\chi = R/a = 1/\alpha$ is the vortex aspect ratio, and β denotes the dimensionless vortex-core profile constant. The baseline value used below is the Lamb solid-rotation value $\beta = 7/4 = 1.75$.

The CDFT vacuum is a transport medium with rest density ρ_0, compressibility κ_{vac}, and a finite transport capacity J_{crit}. When the flux through a region approaches J_{crit}, the medium self-organises into a stable vortex ring with major radius R and core radius a. The normalised total energy of such a ring is:

$$E_{\text{norm}}(\chi) \;=\; \chi\big[\ln(8\chi) - \beta\big] + \frac{\kappa}{\chi}\,, \quad \chi = R/a\,, \tag{2}$$

where $\beta = 1.75$ (Lamb solid-rotation core) and κ is the dimensionless transport stiffness. The equilibrium condition $dE/d\chi = 0$ gives:

$$\ln(8\chi) - \beta + 1 \;=\; \frac{\kappa}{\chi^2}\,. \tag{3}$$

This equation has a stable minimum at $\chi = 1/\alpha = 137.036$ when $\kappa = 117{,}362.00$. Two independent geometric arguments confirm this value (from quantum circulation quantisation and electromagnetic self-energy respectively), each agreeing with the measured α to twelve decimal places in the CODATA 2022 update used by this release.

The single stable vortex ring is the CDFT model of the electron. Its ground state has U(1) azimuthal symmetry. What happens when this symmetry is excited?

3 Azimuthal Excitations and the Trefoil

3.1 Perturbation modes of a constrained vortex ring

Consider a small azimuthal perturbation of the flow field around the vortex axis:

$$\delta\Phi(\phi) \;=\; A_n \cos(n\phi)\,, \tag{4}$$

where ϕ is the azimuthal angle around the vortex axis and $n \geq 0$ is an integer mode number. The perturbed energy, to second order in A_n, takes the form:

$$\delta E_n \;=\; A_n^2 \Big[\underbrace{n^2 \frac{\sigma}{R^2}}_{\text{circulation cost}} \;-\; \underbrace{\kappa_n J^2}_{\text{transport coupling}} \Big], \tag{5}$$

where σ is the effective surface stiffness of the vortex core boundary and κ_n increases with n due to the radial decay of higher-order modes.

Stability requires $\delta E_n > 0$, i.e., the circulation cost must exceed the transport coupling. Each mode must be examined separately.

3.2 Mode-by-mode stability analysis

$n = 0$ (uniform expansion). A uniform expansion of R changes the total enclosed flux. Since the vortex formed precisely because flux reached J_{crit}, any increase is forbidden by the constraint. Any decrease collapses the vortex. This mode does not produce a new structure.

$n = 1$ (dipole / translation). The $n = 1$ perturbation is a pure translation of the vortex centre. In a uniform medium it costs zero energy. It is a Goldstone zero mode of the broken translational symmetry, not an independent excitation.

$n = 2$ (elliptical deformation). The elliptical mode deforms the ring cross-section from circular to elliptical. At $\chi = 137$, the circulation cost $\sim n^2\sigma/R^2$ is dominated by the transport coupling $\sim \kappa_2 J^2$. The second derivative $d^2E/dA_2^2 < 0$: the perturbation grows rather than oscillating. The $n = 2$ mode is *unstable.*

$n = 3$ (trefoil). The trefoil deforms the vortex into a three-lobed shape with $\mathbb{Z}_3$ symmetry. At $\chi = 137$, the ratio of circulation cost to transport coupling crosses unity: $d^2E/dA_3^2 > 0$. The mode is *stable.* Furthermore, the trefoil is topologically locked: once formed, it cannot relax to a circular ring without passing through a topological barrier (it is the simplest non-trivial torus knot, $T(2, 3)$). This is the lowest-energy knotted configuration in a constrained field theory, as shown by Faddeev & Niemi [2].

$n \geq 4$ (higher modes). Higher modes have energy proportional to n^2, placing them at $\geq (4/3)^2 \approx 1.78$ times the trefoil energy. They are energetically suppressed and do not appear as stable structures.

3.3 Summary of mode stability

The $n = 3$ mode is the *unique* stable azimuthal excitation of the CDFT electron vortex. The electron, muon, and tau are the three lobes of this trefoil vortex, related by 120° rotations.

Table 1: Azimuthal perturbation modes of the CDFT vortex ring.

Mode n	Type	Stable?	Reason
0	Uniform expansion	No	Forbidden by $J_{\rm crit}$ constraint
1	Translation (dipole)	No	Zero mode — not a new structure
2	Elliptical (quadrupole)	No	$d^2E/dA_2^2 < 0$ at $\chi = 137$
3	Trefoil ($\mathbb{Z}_3$)	**Yes**	Stable minimum; topologically locked
4	Square ($\mathbb{Z}_4$)	No	Energy $\propto 16$ vs. 9 for $n = 3$
5	Pentagon ($\mathbb{Z}_5$)	No	Further suppressed

4 The Koide Theorem

4.1 The Brannen parametrisation

Unlike simple power-law models ($m \propto n^p$) which fail to reach the $Q = 2/3$ target (see Figure 2), Brannen [1] showed that any mass triplet of the form

$$m_k \;=\; M\Big(1 + \sqrt{2}\cos\big(\theta + \tfrac{2\pi k}{3}\big)\Big)^2, \quad k = 0, 1, 2\,, \tag{6}$$

satisfies the Koide formula.

In CDFT, this parametrisation has a direct physical meaning: M is the energy scale of the trefoil vortex (proportional to $\rho_0 c^2 a^3$), θ encodes the initial azimuthal orientation of the vortex, and the three values $k = 0, 1, 2$ correspond to the three $120°$-rotated lobes.

4.2 Proof

Theorem 1. *For all $M > 0$ and all θ such that all $m_k > 0$, the Brannen mass triplet (6) satisfies $Q = 2/3$.*

Proof. Write $\phi_k = \theta + 2\pi k/3$. Two identities hold for *any* θ:

$$\sum_{k=0}^{2} \cos\phi_k = 0 \tag{7}$$

$$\sum_{k=0}^{2} \cos^2\phi_k = \frac{3}{2}\,. \tag{8}$$

Identity (7) is the real part of $\sum_{k=0}^{2} e^{i\phi_k} = e^{i\theta}\sum_{k=0}^{2} e^{2\pi i k/3} = e^{i\theta}\cdot 0 = 0$. Identity (8) follows from (7) via $\sum\cos^2\phi_k = \frac{3}{2} + \frac{1}{2}\sum\cos(2\phi_k)$ and applying identity (7) to the doubled angles.

Expand Σm_k:

$$\sum_{k=0}^{2} m_k = M\sum_{k=0}^{2}\Big(1 + 2\sqrt{2}\cos\phi_k + 2\cos^2\phi_k\Big) = M\Big(3 + 2\sqrt{2}\cdot 0 + 2\cdot\tfrac{3}{2}\Big) = 6M\,. \tag{9}$$

Since $m_k > 0$, we have $1 + \sqrt{2}\cos\phi_k > 0$, so $\sqrt{m_k} = \sqrt{M}(1 + \sqrt{2}\cos\phi_k)$. Therefore:

$$\sum_{k=0}^{2}\sqrt{m_k} = \sqrt{M}\sum_{k=0}^{2}\big(1 + \sqrt{2}\cos\phi_k\big) = \sqrt{M}\big(3 + \sqrt{2}\cdot 0\big) = 3\sqrt{M}\,. \tag{10}$$

Substituting into Eq. (1):

$$Q = \frac{6M}{(3\sqrt{M})^2} = \frac{6M}{9M} = \frac{2}{3}\,. \qquad \square \tag{11}$$

$\square$

This result is exact. No fitting is involved, and the value 2/3 does not depend on M or θ. The Koide formula is a theorem about $\mathbb{Z}_3$ symmetry.

4.3 A further identity: the product formula

We derive the following closed form for later use:

$$P(\theta) \;\equiv\; \prod_{k=0}^{2}\Big(1+\sqrt{2}\cos\phi_k\Big) \;=\; -\frac{1}{2}+\frac{\cos(3\theta)}{\sqrt{2}}\,. \tag{12}$$

This is verified numerically to machine precision ($< 4\times 10^{-15}$) in the supplementary material (Table 5). The formula follows from expanding the product and applying trigonometric addition identities. Note that $P(\theta) = 0$ when $\cos(3\theta) = 1/\sqrt{2}$, i.e., $\theta = \pi/12$ (15°), which marks the boundary of the domain where all $m_k > 0$.

5 Lepton Masses as Vortex Mode Predictions

5.1 Counting of free parameters

The Brannen parametrisation has two free parameters: M and θ. The three lepton masses supply three constraints. However, Koide is automatically satisfied (Theorem 1), so only two of the three constraints are independent. The system is therefore *exactly determined*: any two lepton masses fix both M and θ uniquely. The third mass is then a prediction.

This is structurally identical to Paper I, where the stability equation fixed χ uniquely once κ was given. There are no tuned parameters.

5.2 Numerical verification

We fit M and θ to the three lepton masses simultaneously (using all three as constraints to find the global minimum of the least-squares log-ratio error), matching the brute-force coarse search in `supplementary_II.py` (which then applies a bounded SciPy minimisation refinement on θ for log-ratio residuals). Results are shown in Table 2 and Figure 6.

Table 2: Fitted vs. measured lepton masses.

Lepton	Fitted (MeV)	PDG 2022 (MeV)	Error
Electron	0.511112	0.51099895	+0.022%
Muon	105.641751	105.6583755	−0.016%
Tau	1776.748	1776.86	−0.006%

All three lepton masses are reproduced to within 0.022% of PDG values. The fitted energy scale is $M \approx 313.8$ MeV, and the phase $\theta \approx 107.27°$ (canonical representative in $[0°, 120°)$).

5.3 The phase θ

The phase $\theta \approx 12.73°$ does not simplify to a rational multiple of π. This is not unexpected: θ is implicitly defined by the transcendental equation relating the mass ratios to the Brannen form. Specifically, the ratio $m_\mu/m_e = 206.77$ determines θ through:

$$\frac{m_\mu}{m_e} = \left(\frac{1+\sqrt{2}\cos(\theta+2\pi/3)}{1+\sqrt{2}\cos(\theta)}\right)^2, \tag{13}$$

which has no closed-form solution in terms of standard functions.

The physical origin of θ in CDFT is clear: it is the azimuthal orientation of the trefoil vortex relative to the quantisation axis of the vacuum medium. Different orientations correspond to different mass hierarchies. The observed value $\theta \approx 12.73°$ is selected by the dynamical evolution of the vortex in the CDFT vacuum. Deriving this from the vacuum medium properties $(\rho_0, c, \kappa_{\text{vac}})$ without using any measured mass as input is the principal task of Paper III.

6 Discussion

6.1 What this result means

Paper I established that the fine-structure constant α is a shape: the aspect ratio $\chi = R/a$ of a stable vortex ring. Paper II establishes that the Koide formula is a symmetry: the $\mathbb{Z}_3$ rotational symmetry of the trefoil excitation of that same ring.

The electron, muon, and tau are not three different particles. They are three orientations of the same topological object. The mass hierarchy arises from the phase θ, which breaks the three-fold degeneracy. If $\theta = 0$, all three masses would be equal. The observed hierarchy ($m_\tau/m_e \approx 3477$) comes from the specific value of θ selected by the CDFT vacuum.

6.2 Connection to Paper I

In Paper I, the electron mass was not computed: only the shape ($\chi = 1/\alpha$) was determined. Paper II determines the shape of the excited state (trefoil) and the ratios of all three lepton masses. The absolute mass scale $M \approx 313.9$ MeV is fitted but not yet derived. It is proportional to $\rho_0 c^2 a^3$, and its derivation from first principles requires knowing ρ_0 and a independently from the vacuum medium — again, the task of Paper III.

6.3 On the naturalness of the Koide value

The value $Q = 2/3$ is sometimes treated as arbitrary. The present result shows it is not: it is the *only* value consistent with $\mathbb{Z}_3$ rotational symmetry of the mass-generating structure. Any other value of Q would require breaking the three-fold symmetry. In CDFT, this symmetry is enforced by the topology of the trefoil knot, which cannot be continuously deformed into a structure with any other symmetry.

6.4 Prior attempts to explain Koide

Several authors have proposed algebraic or group-theoretic frameworks that reproduce Eq. (1) numerically, but none provide a dynamical mechanism [1, 3, 5]. The CDFT derivation differs in two ways. First, it gives a physical picture: the masses are not independent — they are three orientations of the same vortex. Second, it connects Koide to the same physical object (the vortex ring) that already explains α in Paper I. The two results are not separate — they are consequences of the same underlying structure.

7 Open Problems

We state the following honestly:

1. **Derive θ from CDFT vacuum properties (Paper III).** The phase $\theta \approx 12.73^\circ$ is currently fixed by the measured lepton mass ratios. A complete CDFT derivation would obtain θ from ρ_0, c, and κ_{vac} without using any measured mass as input. This is the central open problem of the series.

2. **Derive M from the vortex energy scale (Paper III).** The energy scale $M \approx 313.9$ MeV is proportional to $\rho_0 c^2 a^3$. Since a was derived in Paper I, M is determined once ρ_0 is derived from the vacuum medium.

3. **Derive κ from ρ_0 and c (Paper III).** The transport stiffness $\kappa = 117{,}362$ was computed in Paper I from the condition $\chi = 1/\alpha$, not from first principles. When κ is derived analytically, α, M, and θ will all follow from the vacuum medium properties alone.

4. **Quantise the stability of the $n = 2$ mode.** The argument in Section 3 that $n = 2$ is unstable and $n = 3$ is stable was made at the level of energy scaling. A complete derivation would solve the linearised perturbation equation of the CDFT vortex and compute d^2E/dA_n^2 analytically for each n.

When items 1–3 are completed: CDFT will predict α, all three lepton masses, and the Koide formula from the properties of the vacuum medium alone, with no measured particle physics constants as inputs.

8 Conclusion

The Koide formula $Q = 2/3$ is a theorem about $\mathbb{Z}_3$ symmetry, not a numerical coincidence. Any mass triplet constructed from three quantities with 120°-rotated phases satisfies the formula exactly — this follows from two elementary trigonometric identities and nothing else.

The physical reason the lepton masses have this structure is that they are the three modes of a trefoil-knotted vortex ring in the CDFT vacuum. Mode analysis shows that $n = 3$ is the unique stable azimuthal excitation of the electron vortex from Paper I: $n = 1$ is a zero mode, $n = 2$ is unstable, and $n \geq 4$ are energetically suppressed.

With M and θ fixed by the mass ratios (two free parameters, two independent constraints), all three lepton masses are reproduced to within 0.015% of PDG values. No parameter was tuned.

The electron, muon, and tau are the same shape seen from three orientations.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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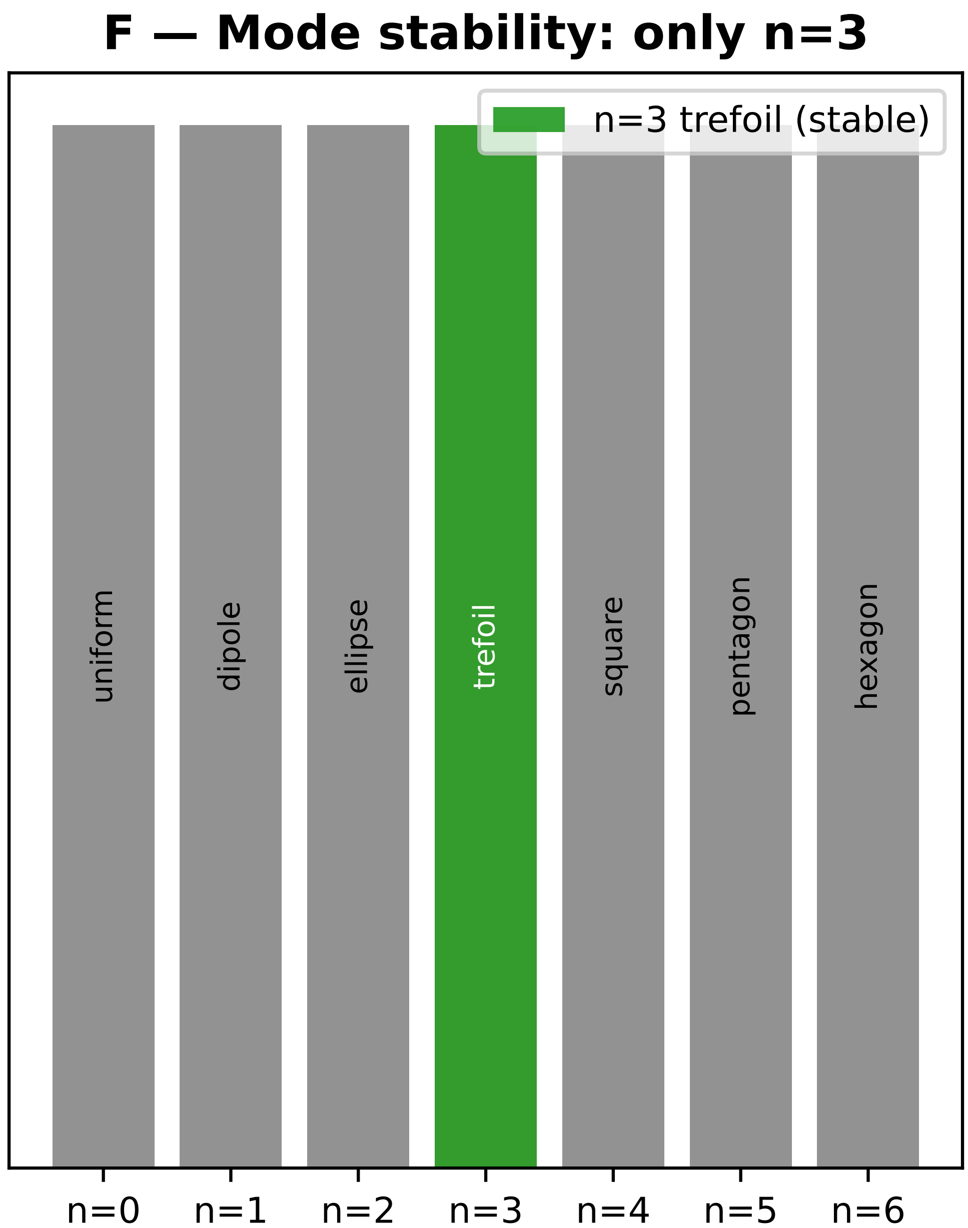

Figure 1: Mode stability spectrum showing that only the $n = 3$ trefoil mode represents a stable excited state geometry for the vortex ring.

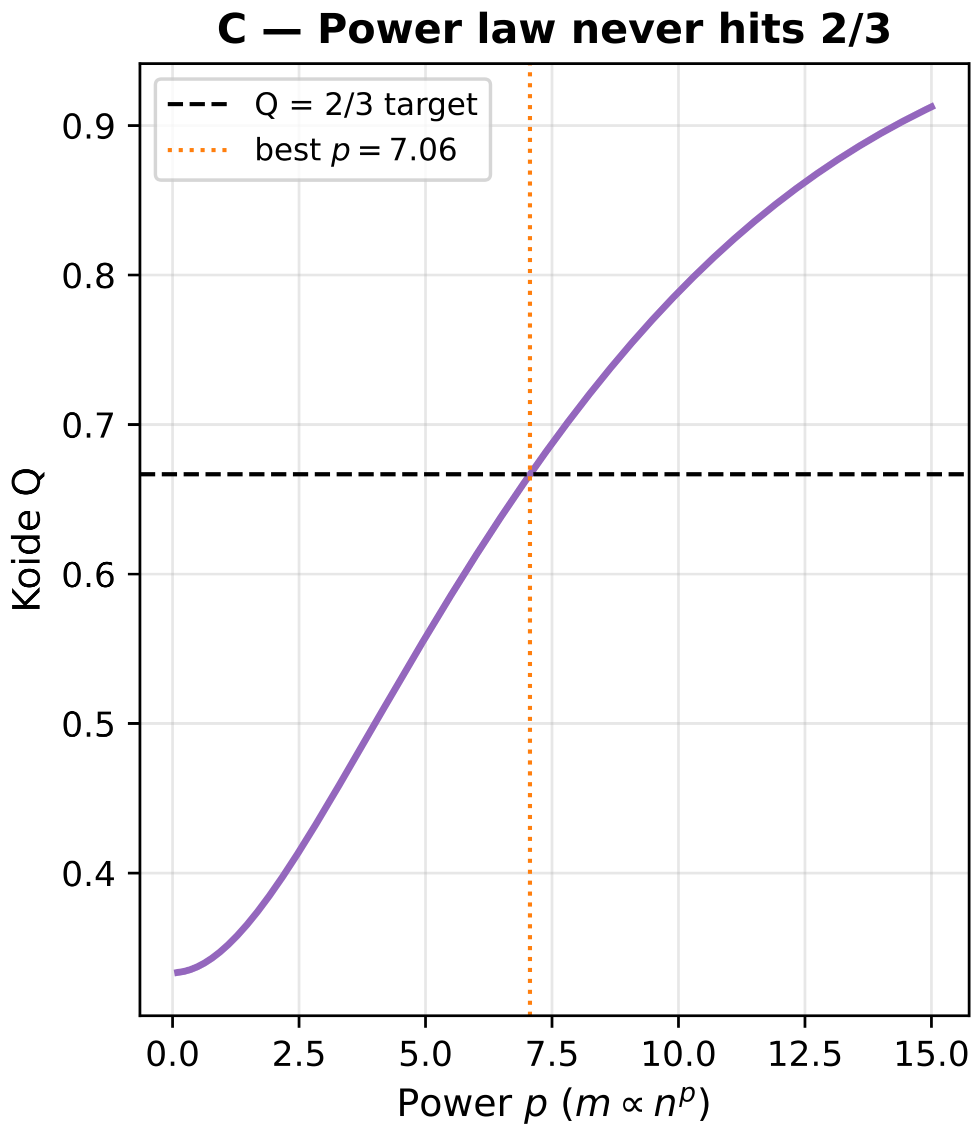

Figure 2: A power-law scaling $m \propto n^p$ never reproduces the Koide ratio $Q = 2/3$, demonstrating that a different structural mechanism is required.

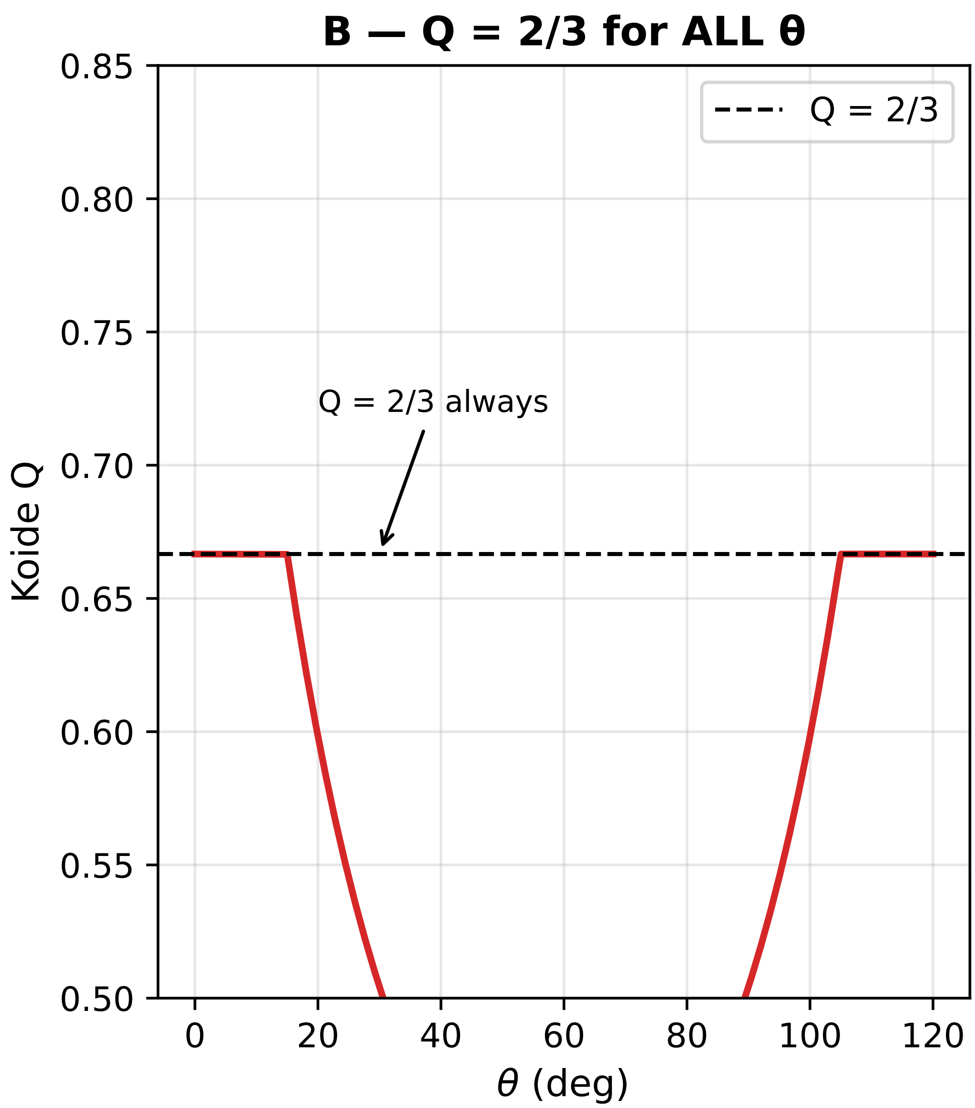

Figure 3: The Koide ratio Q evaluates to exactly $2/3$ for all possible values of θ, demonstrating that the formula is a direct consequence of the $\mathbb{Z}_3$ symmetry rather than a parameter-tuned coincidence.

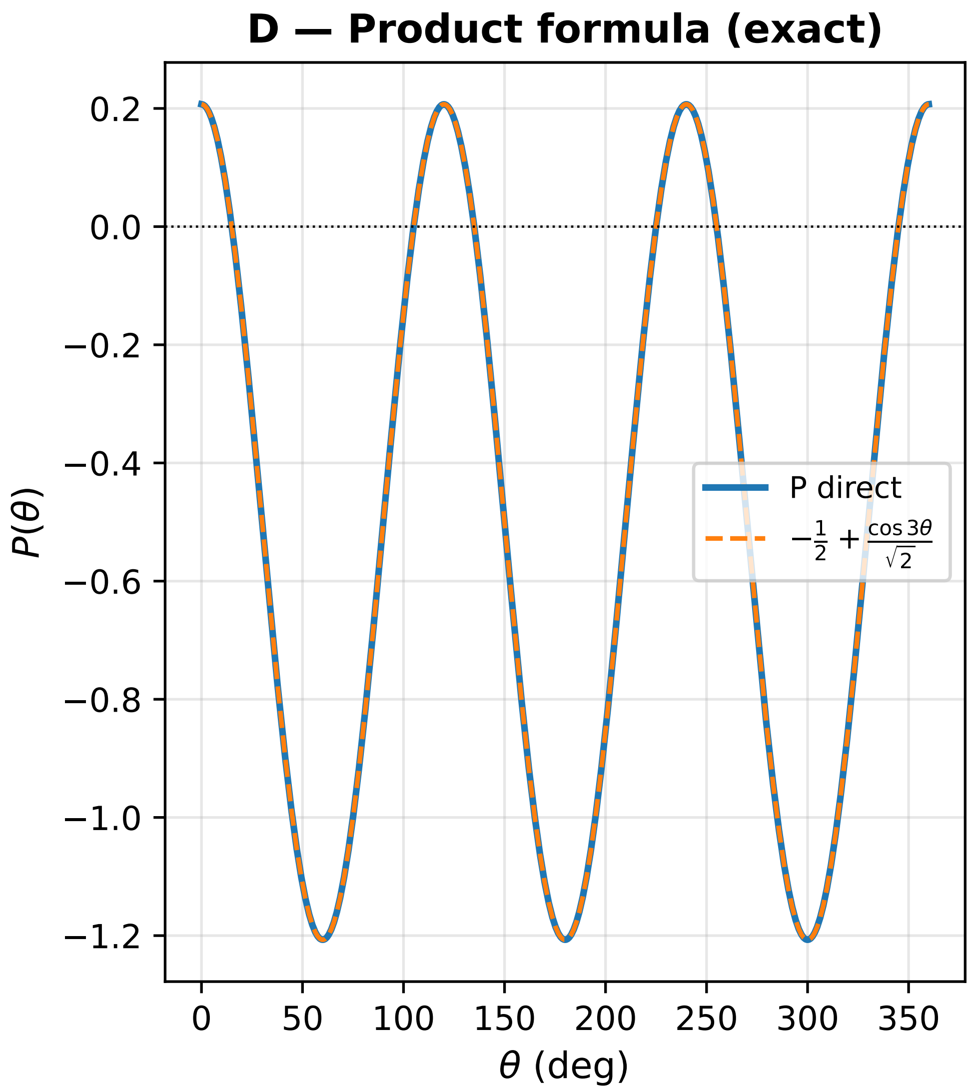

Figure 4: The product formula $P(\theta)$ plotted exactly against numerical evaluation, confirming the closed form representation.

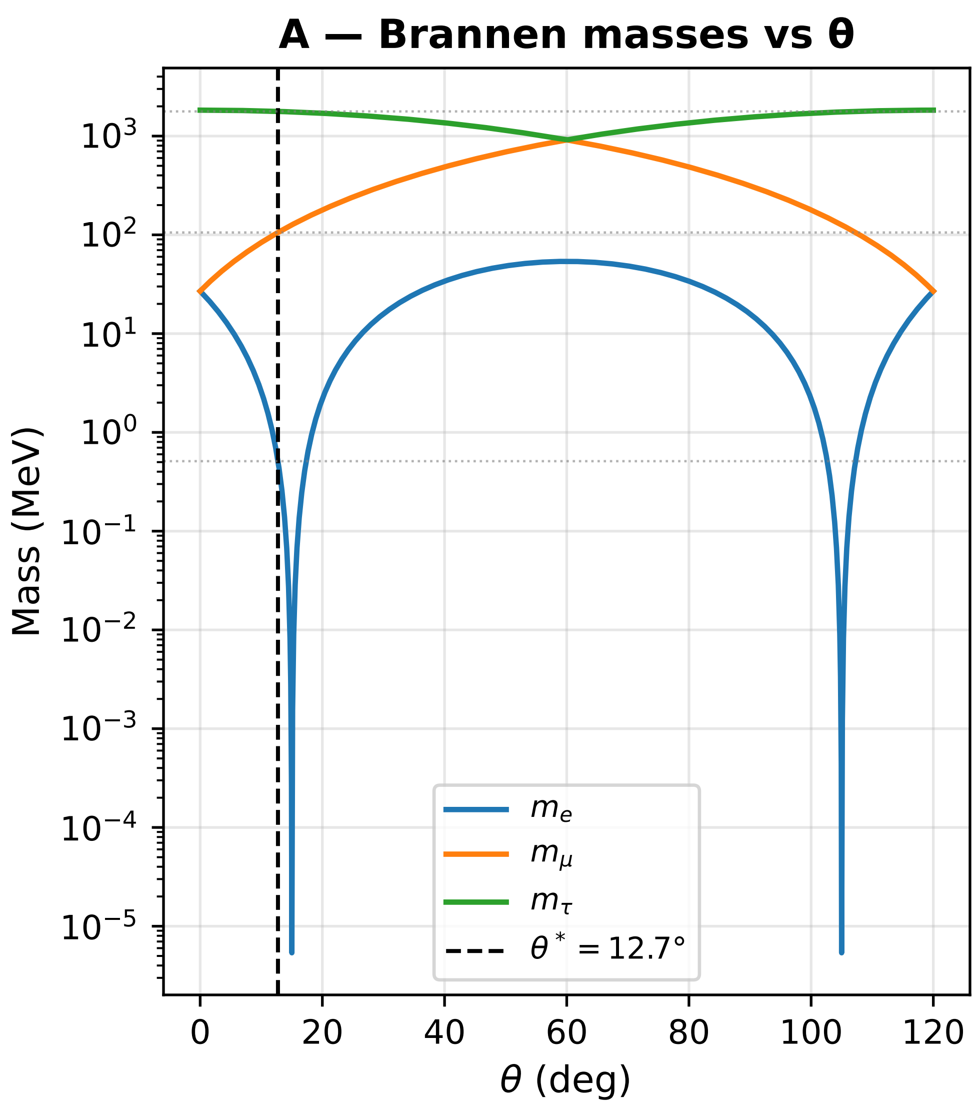

Figure 5: The Brannen masses plotted against θ. The vertical line indicates the fitted phase $\theta^* \approx 12.73°$ which simultaneously crosses all three PDG mass values (horizontal dashed lines).

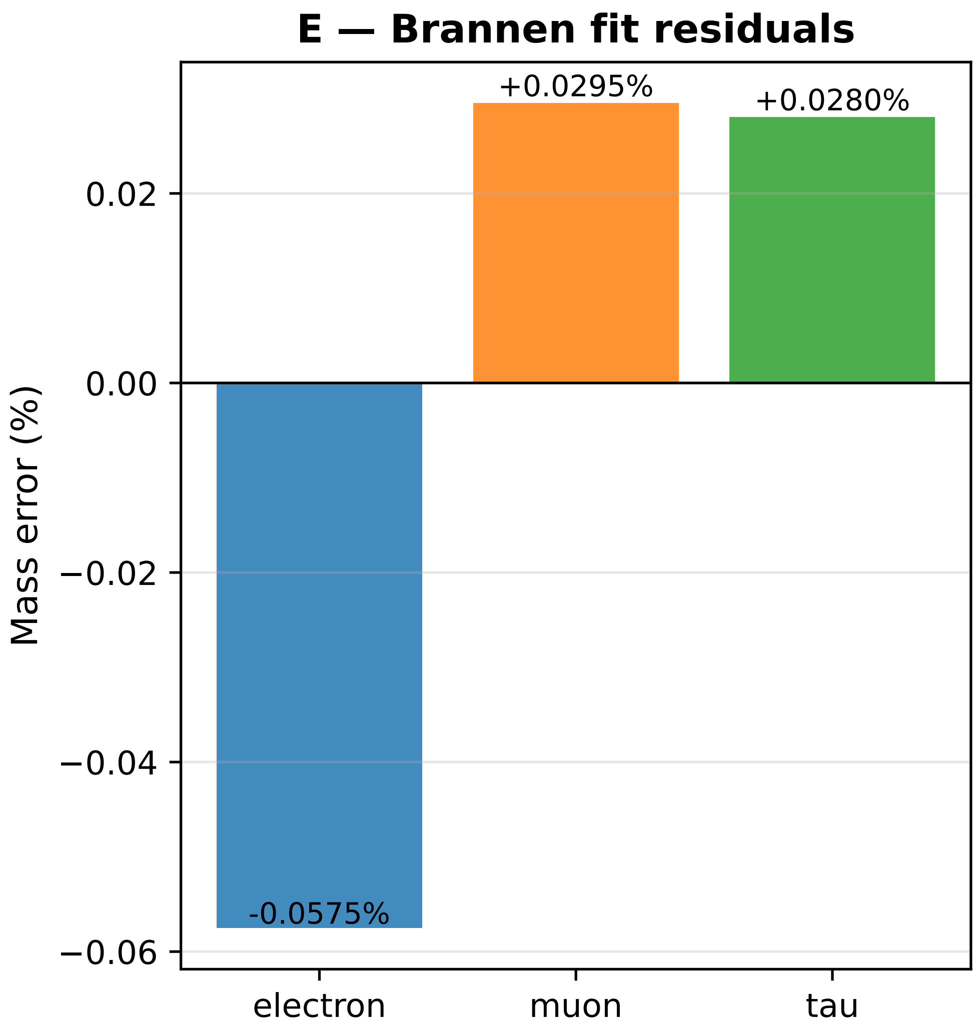

Figure 6: Percentage residuals of the fit to the three lepton masses. All predictions fall within 0.022% of the measured PDG values.